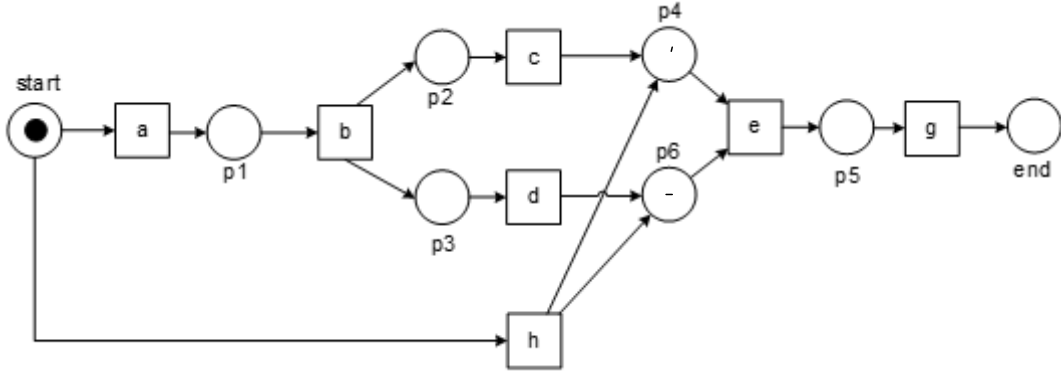


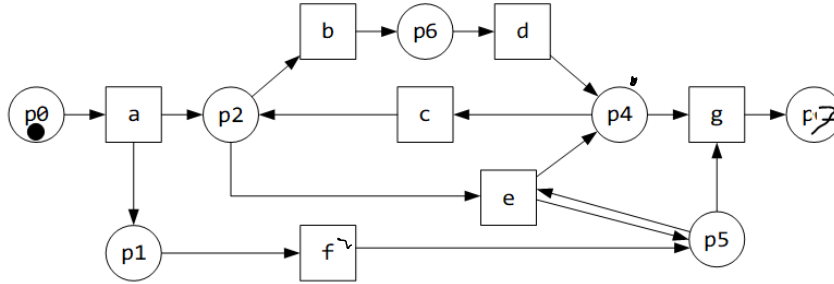
2 Conformance Checking

Exercise 2.1 Consider the following model:



Moreover, consider the following event log: $L = [\langle a, g \rangle^3, \langle h, e, g \rangle^3]$. Compute fitness for the given event log and the given model using *token-based fitness*.

Exercise 2.2 Consider event log $L = [\langle a, b, e, d, g \rangle^{10}, \langle a, b, f, d, g \rangle^4]$ and the following Petri net:



1. Compute the fitness for the given log on the given model using token-based fitness.
2. Illustrate an example of a potential trace where the number of tokens that is consumed is not equal to the number of tokens that are produced.
3. For the following trace, select the appropriate optimal alignment (a), (b) or (c). Assume the default cost function assigning cost 1 to each deviation. $\langle a, c, e, f, e, g \rangle$

(a)	$\frac{\text{log trace}}{\text{execution sequence}}$	$\begin{array}{ c c c c c c c c } \hline a & \gg & \gg & c & e & f & e & g \\ \hline a & b & d & c & \gg & f & e & g \\ \hline \end{array}$
-----	--	---

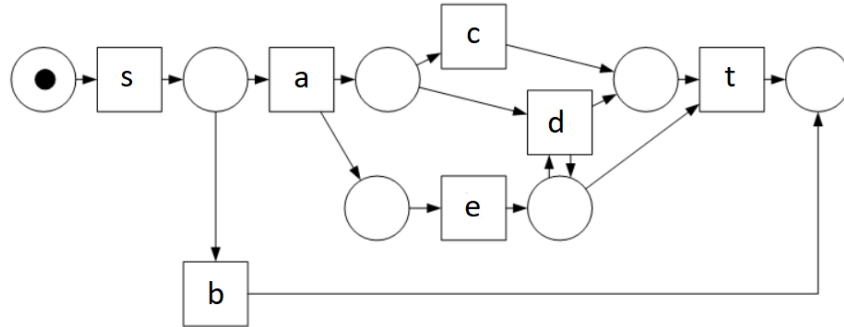
(b)	$\frac{\text{log trace}}{\text{execution sequence}}$	$\begin{array}{ c c c c c c } \hline a & c & e & f & e & g \\ \hline a & \gg & \gg & f & e & g \\ \hline \end{array}$
-----	--	---

(c)	$\frac{\text{log trace}}{\text{execution sequence}}$	$\begin{array}{ c c c c c c c } \hline a & \gg & \gg & e & f & e & g \\ \hline a & b & d & e & f & e & g \\ \hline \end{array}$
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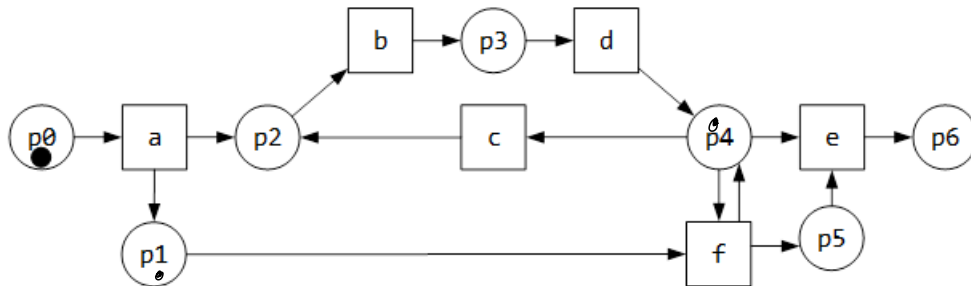
4. For the following trace, draw the synchronous product and use it to find the appropriate optimal alignment. Assume the default cost function assigning cost 1 to each deviation. $\langle a, b, d, c, e, g \rangle$

Exercise 2.3 Consider event log $L = [\langle s, a, c, e, t \rangle^5, \langle a, e, d, t \rangle^3, \langle s, a, e, t \rangle^4, \langle s, b \rangle^{10}]$ and the following Petri net:



1. Compute fitness of the event log with respect to the model using token-based fitness.
2. Give an optimal alignment for the trace $\langle a, b, d, c, d, e, t \rangle$, and, its corresponding fitness. Use the standard cost function, i.e. log/model moves have cost 1, synchronous moves have value 0.

Exercise 2.4 Consider event log $L = [\langle a, b, e, d, c \rangle^{10}, \langle a, b, d, f, e \rangle^5]$ and the following Petri net:



1. Compute the fitness for the given log on the given model using token-based fitness.
2. Illustrate an example of a potential trace where the number of tokens that is consumed is not equal to the number of tokens that are produced.
3. For the following trace, select the appropriate optimal alignment (a), (b) or (c). Assume the default cost function assigning cost 1 to each deviation. $\langle a, c, e, f, e, g \rangle$

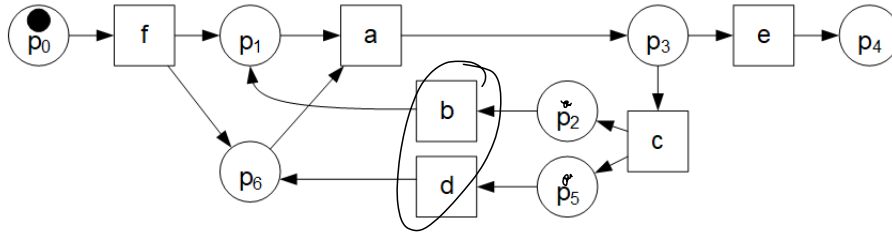
(a)	log trace	a	>>	>>	c	e	f	e	g
	execution sequence	a	b	d	>>	>>	f	e	>>

(b)	log trace	a	c	e	f	e	g
	execution sequence	a	c	>>	f	e	>>

(c)	log trace	a	>>	>>	c	e	>>	>>	f	e	g
	execution sequence	a	b	d	c	>>	b	d	f	e	>>

4. For the following trace, provide the synchronous product and use it to compute an optimal alignment. Assume the default cost function assigning cost 1 to each deviation. $\langle a, b, f, c, d, e \rangle$

Exercise 2.5 Consider event log $L = [\langle f, a, b, d, a \rangle^{10}, \langle f, a, c, b, e \rangle^5]$ and the following Petri net:



1. Compute the fitness for the given log on the given model using token-based fitness.
2. Illustrate an example of a potential trace where the number of tokens that is consumed is not equal to the number of tokens that are produced.
3. For the following trace, select the appropriate optimal alignment (a), (b) or (c). Assume the default cost function assigning cost 1 to each deviation. $\langle f, a, c, a, e \rangle$

(a)

<i>log trace</i>	f	a	c	$\gg$	$\gg$	a	e
<i>execution sequence</i>	f	a	c	b	d	a	e

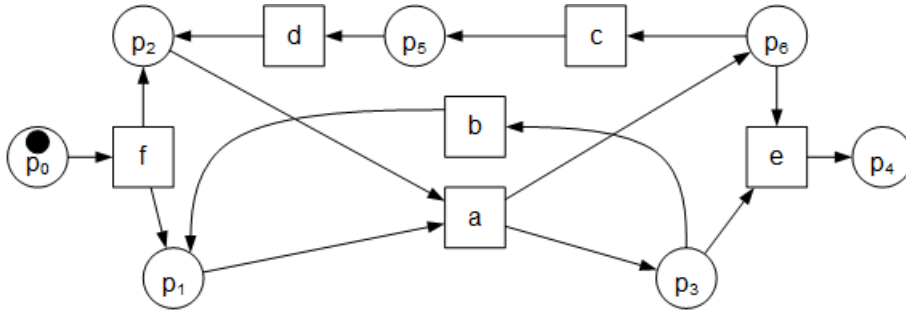
(b)

<i>log trace</i>	f	a	c	$\gg$	a	e
<i>execution sequence</i>	f	a	c	d	a	e

(c)

<i>log trace</i>	f	a	c	a	e
<i>execution sequence</i>	f	$\gg$	$\gg$	$\gg$	e

4. For the following trace, compute the synchronous product and use it to compute an optimal alignment. Assume the default cost function assigning cost 1 to each deviation. $\langle a, d, b, e, g \rangle$



1. Compute the fitness for the given log on the given model using token-based fitness.
2. Illustrate an example of a potential trace where the number of tokens that is consumed is not equal to the number of tokens that are produced.
3. For the following trace, select the appropriate optimal alignment (a), (b) or (c). Assume the default cost function assigning cost 1 to each deviation. $\langle f, a, c, e \rangle$

(a)

<i>log trace</i>	f	a	c	$\gg$	$\gg$	$\gg$	e
<i>execution sequence</i>	f	a	c	b	d	a	e

(b)

<i>log trace</i>	f	a	c	$\gg$	e
<i>execution sequence</i>	f	a	c	d	e

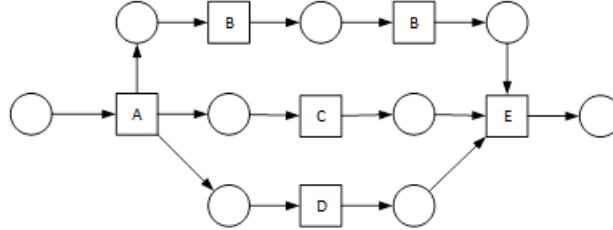
(c)

<i>log trace</i>	f	a	c	e
<i>execution sequence</i>	f	a	$\gg$	e

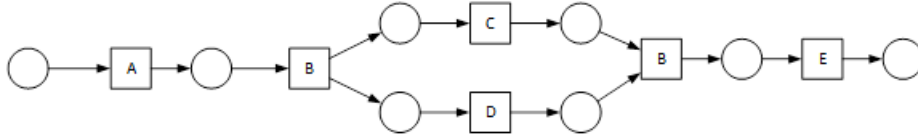


4. For the following trace, provide the synchronous product and use it to compute an optimal alignment. Assume the default cost function assigning cost 1 to each deviation.
 $\langle a, c, d, a, b, a, e \rangle$

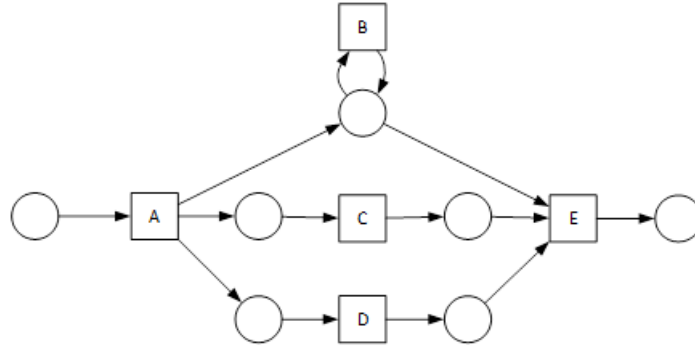
Exercise 2.7 Consider event log $L = [\langle a, b, c, d, b, e \rangle^{200}, \langle a, b, d, c, b, e \rangle^{100}, \langle a, d, b, c, e \rangle^2, \langle a, d, b, b, c, e \rangle^2]$ and the following Petri nets:



(A)



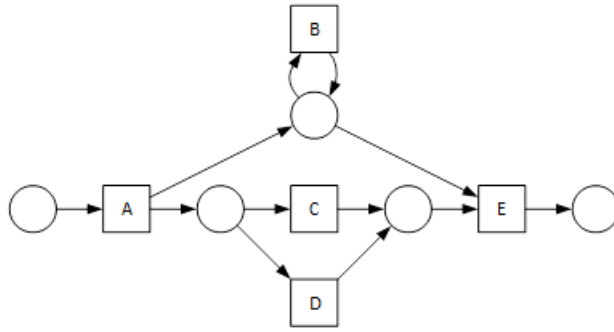
(B)



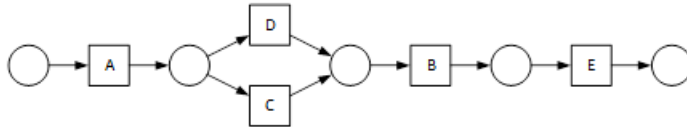
(C)

Rank these models from most fitting to least fitting with respect to log L .

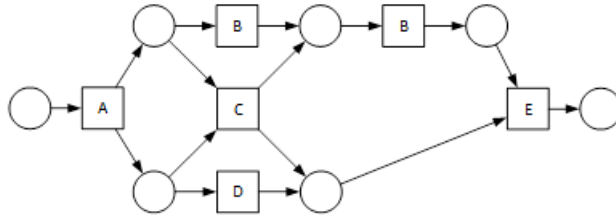
Exercise 2.8 Consider event log $L = [\langle a, b, d, b, e \rangle^{200}, \langle a, c, b, e \rangle^{100}, \langle a, b, b, d, e \rangle^3, \langle a, b, c, e \rangle^4]$ and the following Petri nets:



(A)



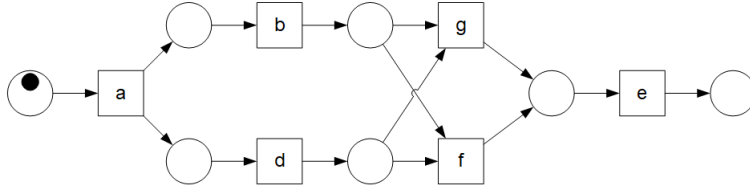
(B)



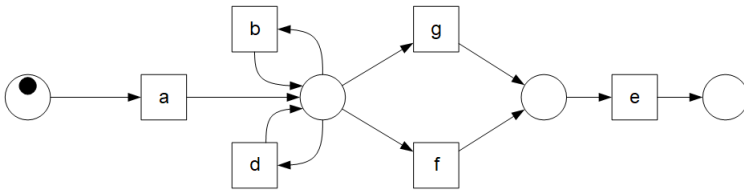
(C)

Rank these models from most fitting to least fitting with respect to $\log L$.

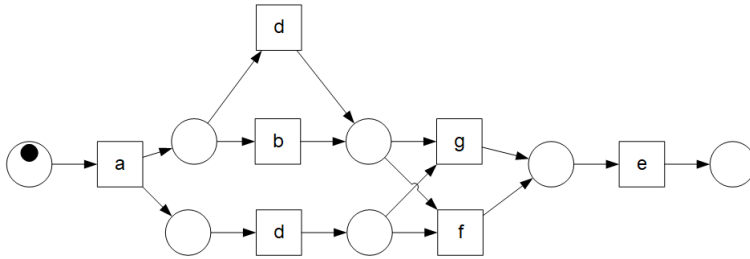
Exercise 2.9 Consider event log $L = [\langle a, b, d, f, e \rangle^{200}, \langle a, d, b, g, e \rangle^{50}, \langle a, d, d, g, e \rangle^5, \langle a, g, e \rangle^2]$ and the following Petri nets:



(A)



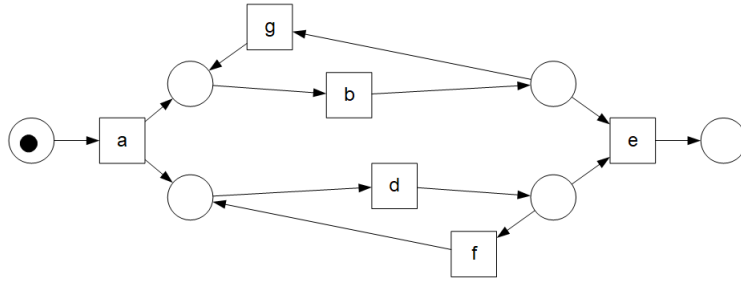
(B)



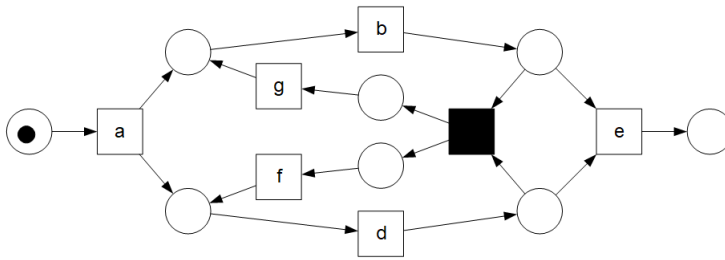
(C)

Rank these models from most fitting to least fitting with respect to $\log L$.

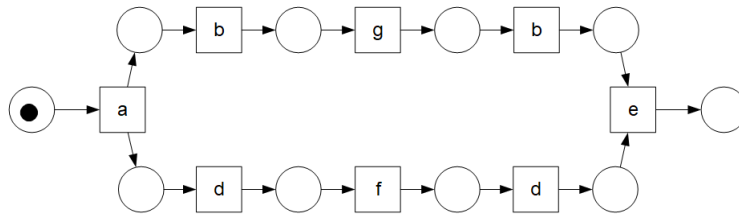
Exercise 2.10 Consider event log $L = [\langle a, b, d, f, e \rangle^{200}, \langle a, d, b, g, e \rangle^{50}, \langle a, d, d, g, e \rangle^5, \langle a, g, e \rangle^2]$ and the following Petri nets (note that one model contains an ‘invisible’ transition, i.e. a transition which does not correspond to an activity logged in the event log):



(A)



(B)



(C)

Rank these models from most fitting to least fitting with respect to $\log L$.

Solutions

Solution 2.1

The table for token-based fitness is depicted below:

Token based fitness:

Trace	Freq	P	R	C	M	P (all)	R (all)	C (all)	M (all)
$\langle a, g \rangle$	3	3	$1^{(p_1)}$	3	$1^{(p_5)}$	9	3	9	3
$\langle h, e, g \rangle$	3	4	0	5	0	50	0	50	0
total						59	3	59	3

$$\text{fitness} = \frac{1}{2}(1 - 3/59) + \frac{1}{2}(1 - 3/59) = 0.9492$$

Solution 2.2

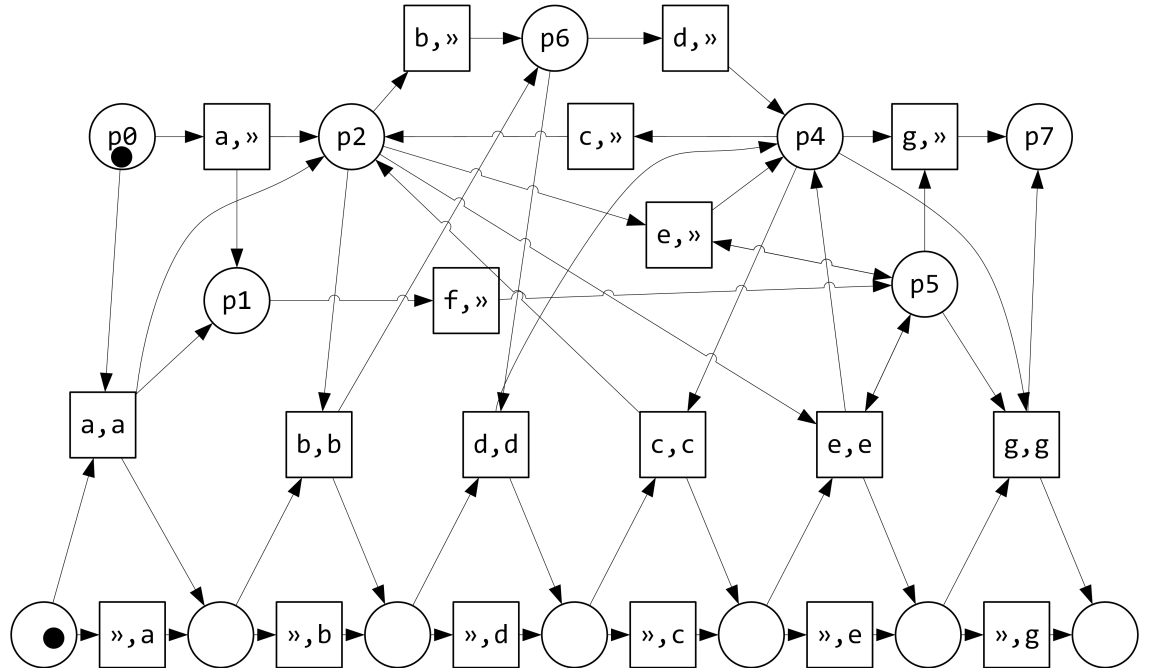
1. The table for token-based fitness is depicted below:

Token based fitness:

Trace	Freq	P	R	C	M	P (all)	R (all)	C (all)	M (all)
$\langle a, b, e, d, g \rangle$	10	8	$2^{(p_1, p_4)}$	8	$2^{(p_2, p_5)}$	80	20	80	20
$\langle a, b, f, d, g \rangle$	4	7	0	7	0	28	0	28	0
total						108	20	108	20

$$\text{fitness} = \frac{1}{2}(1 - 20/108) + \frac{1}{2}(1 - 20/108) = \frac{88}{108} = 0.8148$$

2. An example of such a trace is $\langle a \rangle$.
3. b is the correct alignment.
4. Below is the synchronous product



An optimal alignment with cost 1 is:

<i>log trace</i>	a	b	d	c	»	e	g
<i>execution sequence</i>	a	b	d	c	f	e	g

Solution 2.3

1.

Token based fitness:

Trace	Freq	P	R	C	M	P (all)	R (all)	C (all)	M (all)
s, a, c, e, t	5	7	0	7	0	35	0	35	0
a, e, d, t	3	7	1	7	1	21	3	21	3
s, a, e, t	4	6	1	6	1	24	4	24	4
s, b	10	3	0	3	0	30	0	30	0
total						110	7	110	7

$$\text{fitness} = \frac{1}{2}(1 - 7/110) + \frac{1}{2}(1 - 7/110) = 0.9364$$

2.

<i>log trace</i>	»	a	b	d	c	d	e	t
<i>execution sequence</i>	s	a	»	»	c	»	e	t

Cost of alignment: 4

Cost of “best of worst” alignment: $2 + 7 = 9$

$$\text{Fitness} = 1 - \left(\frac{4}{9}\right) = 5/9 = 0.55$$

Solution 2.4

1. The table for token-based fitness is depicted below:

Token based fitness:

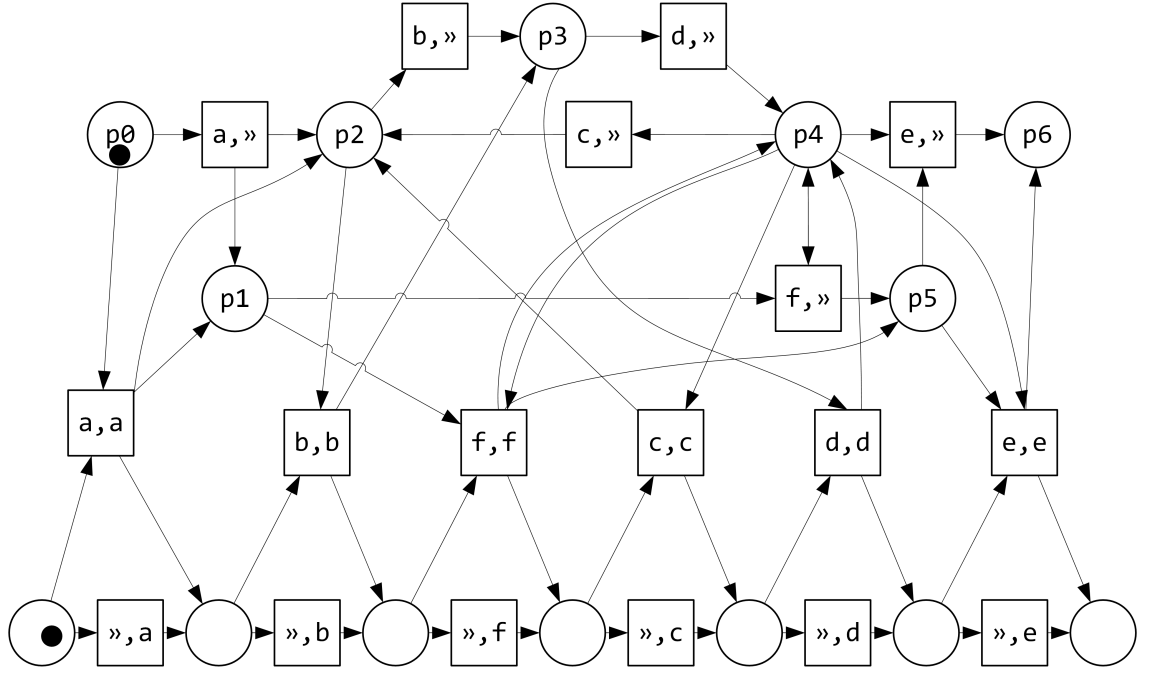
Trace	Freq	per trace				per trace class			
		P	C	R	M	P	C	R	M
$\langle a, b, e, d, c \rangle$	10	1,2,1,1,1,1	1,1,2,1,1,1	p_1, p_2	p_4, p_5	70	70	20	20
$\langle a, b, d, f, e \rangle$	5	1,2,1,1,2,1	1,1,1,2,2,1	0	0	40	40	0	0
total						110	110	20	20

$$\text{fitness} = \frac{1}{2}\left(1 - \frac{M}{C}\right) + \frac{1}{2}\left(1 - \frac{R}{P}\right) = \frac{1}{2}\left(1 - 20/110\right) + \frac{1}{2}\left(1 - 20/110\right) = \frac{90}{110} = 0.8182$$

2. An example of such a trace is $\langle a \rangle$.

3. a is the correct alignment.

4. The synchronous product looks as follows:



And an optimal alignment with cost 2 is:

$\log \text{ trace}$	a	b	$\gg$	f	c	$\gg$	d	e
execution sequence	a	b	d	f	c	b	d	e

Solution 2.5

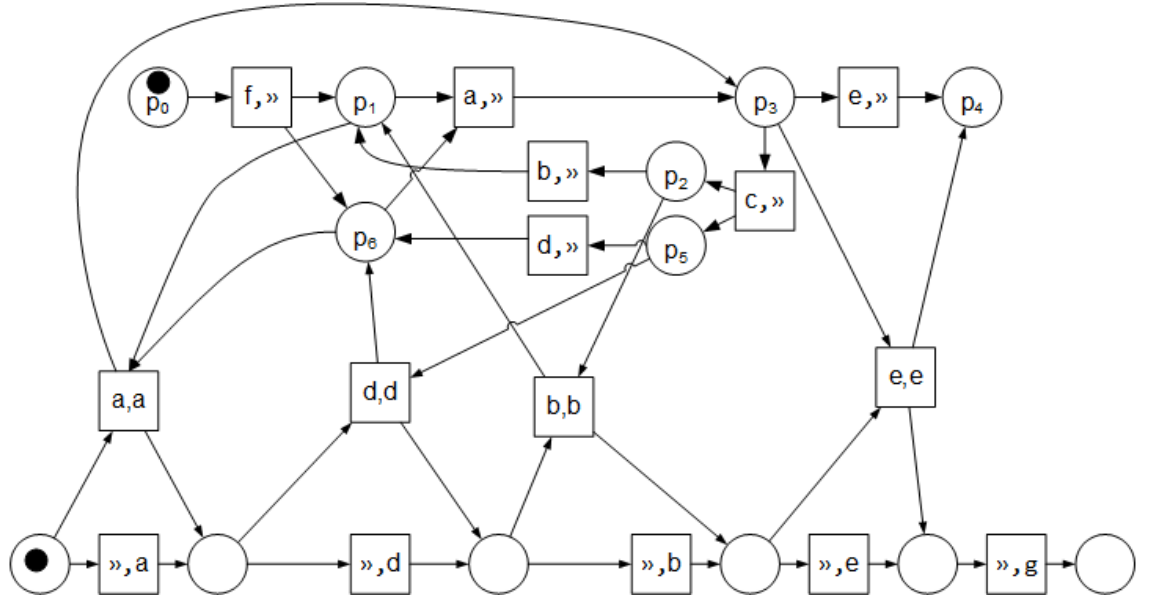
1. The table for token-based fitness is depicted below:

Token based fitness:

Trace	Freq	per trace				per trace class			
		P	C	R	M	P	C	R	M
$\langle f, a, b, d, a \rangle$	10	7	8	2	3	70	80	20	30
$\langle f, a, c, b, e \rangle$	5	8	7	2	1	40	35	10	5
total						110	115	30	35

$$\text{fitness} = \frac{1}{2}(1 - \frac{M}{C}) + \frac{1}{2}(1 - \frac{R}{P}) = \frac{1}{2}(1 - 35/115) + \frac{1}{2}(1 - 30/110) = \frac{180}{253} = 0.7115$$

2. An example of such a trace is $\langle f \rangle$.
3. a is the correct alignment.
4. The synchronous product is depicted below (note the extra transition “g” which does not appear in the model)



An optimal alignment, with cost 4 is:

<i>log trace</i>	»	a	»	d	b	»	e	g
<i>execution sequence</i>	f	a	c	d	b	a	e	»

Or, equally optimal:

<i>log trace</i>	»	a	d	b	e	g
<i>execution sequence</i>	f	a	»	»	e	»

Solution 2.6

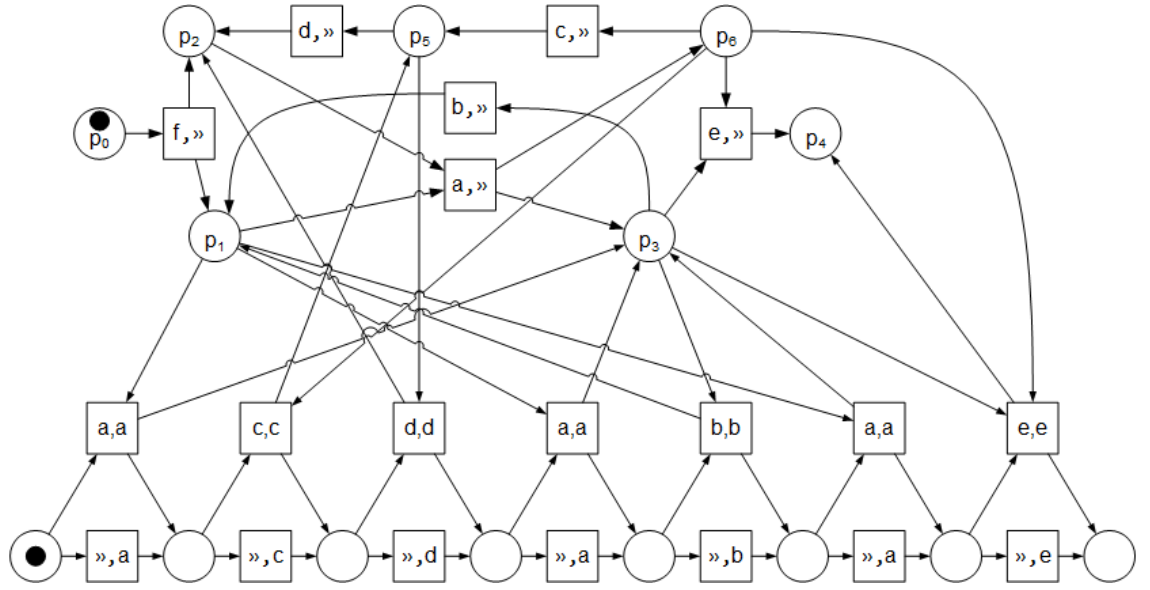
- The table for token-based fitness is depicted below:

Token based fitness:

Trace	Freq	per trace				per trace class			
		P	C	R	M	P	C	R	M
$\langle f, a, b, c, d, a, e \rangle$	10	11	11	0	0	110	110	0	0
$\langle f, a, c, b, a \rangle$	5	9	8	3	2	45	40	15	10
total						155	150	15	10

$$\text{fitness} = \frac{1}{2}(1 - \frac{M}{C}) + \frac{1}{2}(1 - \frac{R}{P}) = \frac{1}{2}(1 - 10/150) + \frac{1}{2}(1 - 15/155) = \frac{427}{465} = 0.9183$$

- An example of such a trace is $\langle f \rangle$.
- c is the correct alignment.
- The synchronous product is depicted below:



An optimal alignment with cost 2 is shown below:

<i>log trace</i>	»	a	c	d	a	b	a	e
<i>execution sequence</i>	f	a	c	d	»	b	a	e

Solution 2.7

From most fitting to least fitting, the models are C, A, B.

Solution 2.8

From most fitting to least fitting, the models are A, C, B.

Solution 2.9

From most fitting to least fitting, the models are B, C, A.

Solution 2.10

From most fitting to least fitting, the models are A, B, C.